

Docker Volumes Lab Handbook

From Scratch - Understanding Persistent Storage with Docker Volumes

Learning Objectives

- Understand why containers lose data
- Understand persistent storage
- Create Docker volumes
- Attach volumes to containers
- Persist PostgreSQL data
- Understand named volumes and bind mounts
- Troubleshoot volume issues

Business Scenario

- Customer data must survive container deletion and recreation.

Lab 1 - Data Loss Demo

- Run PostgreSQL without a volume
- Create sample data
- Delete container
- Observe data loss

Lab 2 - Create a Docker Volume

- `docker volume create postgres-data`
- `docker volume ls`

Lab 3 - Attach Volume to PostgreSQL

- `docker run -d --name postgres-db -v postgres-data:/var/lib/postgresql/data -e POSTGRES_PASSWORD=postgres postgres:16`

Lab 4 - Verify Persistence

- Insert records
- Delete container
- Recreate container using same volume
- Verify records still exist

Named Volumes vs Bind Mounts

- Named Volume = Docker managed
- Bind Mount = Host folder mapping

Verification Commands

- `docker volume ls`

- `docker volume inspect postgres-data`
- `docker inspect postgres-db`

Golden Rule

- Containers are temporary
- Data is permanent
- Use volumes for databases, logs and uploads